E-Commerce Database Management System

Week 3: Seller and Inventory Management System

1. Seller and Inventory Table Design

Week 3 adds inventory tracking to the database. The seller table from Task 1 already holds the supplier details, so the main addition this week is an inventory table. It records how many units of each product a seller has in stock, a stock status, and the time of the last change.

Week 3 objectives

- Use the seller table and create the inventory table, with the Seller to Product and Product to Inventory relationships.
- Apply primary key, foreign key, NOT NULL, UNIQUE and CHECK constraints.
- Add sellers, assign products to sellers, list and count the products of each seller, and update seller details.
- Run inventory queries: available products, out-of-stock products, stock below 10, stock update and deletion of discontinued inventory.
- Generate seller-wise and inventory analysis reports.

How Week 3 builds on earlier tasks. The seller table from Task 1 already has the Week 3 columns, and product.seller_id already links each product to its seller. This week a CHECK rule is added so that a phone number must have exactly 10 digits, and the new inventory table is created. Three sellers and six products are added, the new products are assigned to sellers, and every product receives an inventory record. Stock appears in both the product table (Task 2) and the inventory table, so each stock update changes both tables in one statement.

1.1 Seller (table: seller)

Attribute	Data type	Constraint	Description
seller_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID of the seller
seller_name	VARCHAR(100)	NOT NULL	Business name of the seller
email	VARCHAR(100)	NOT NULL, UNIQUE	Contact email
phone	VARCHAR(15)	NOT NULL, UNIQUE, CHECK (exactly 10 digits)	Contact number
address	VARCHAR(255)	NOT NULL	Business address

1.2 Inventory (table: inventory)

Attribute	Data type	Constraint	Description
inventory_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID of the inventory record
product_id	INT	NOT NULL, UNIQUE, FOREIGN KEY references product(product_id), ON DELETE CASCADE	Product being tracked (one record per product)
seller_id	INT	NOT NULL, FOREIGN KEY references seller(seller_id)	Seller who supplies the product
stock_quantity	INT	NOT NULL, DEFAULT 0, CHECK (stock_quantity >= 0)	Units currently in stock
stock_status	VARCHAR(20)	DEFAULT 'In Stock'	Stock level label: In Stock, Low Stock or Out of Stock
last_updated	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP, ON UPDATE CURRENT_TIMESTAMP	Date and time of the last change to the record

1.3 Stock Status Rule

The stock_status column holds a label that follows this rule. It is set when the inventory record is created and refreshed with one UPDATE statement after every stock change.

Stock quantity	Stock status
0	Out of Stock
1 to 9	Low Stock
10 or more	In Stock

1.4 Constraint Summary

Constraint	Purpose	Where it is used
PRIMARY KEY	Identifies each row uniquely	seller.seller_id, inventory.inventory_id

Constraint	Purpose	Where it is used
FOREIGN KEY	Links a row to a row in another table	product.seller_id to seller, inventory.product_id to product, inventory.seller_id to seller
NOT NULL	Makes a value compulsory	Seller name, email, phone and address, inventory product_id, seller_id and stock_quantity
UNIQUE	Prevents duplicate values	Seller email and phone, inventory.product_id
CHECK	Rejects invalid values	seller.phone has exactly 10 digits, inventory.stock_quantity >= 0
DEFAULT	Supplies a value when none is given	stock_quantity (0), stock_status (In Stock), last_updated (current time)

1.5 Relationships

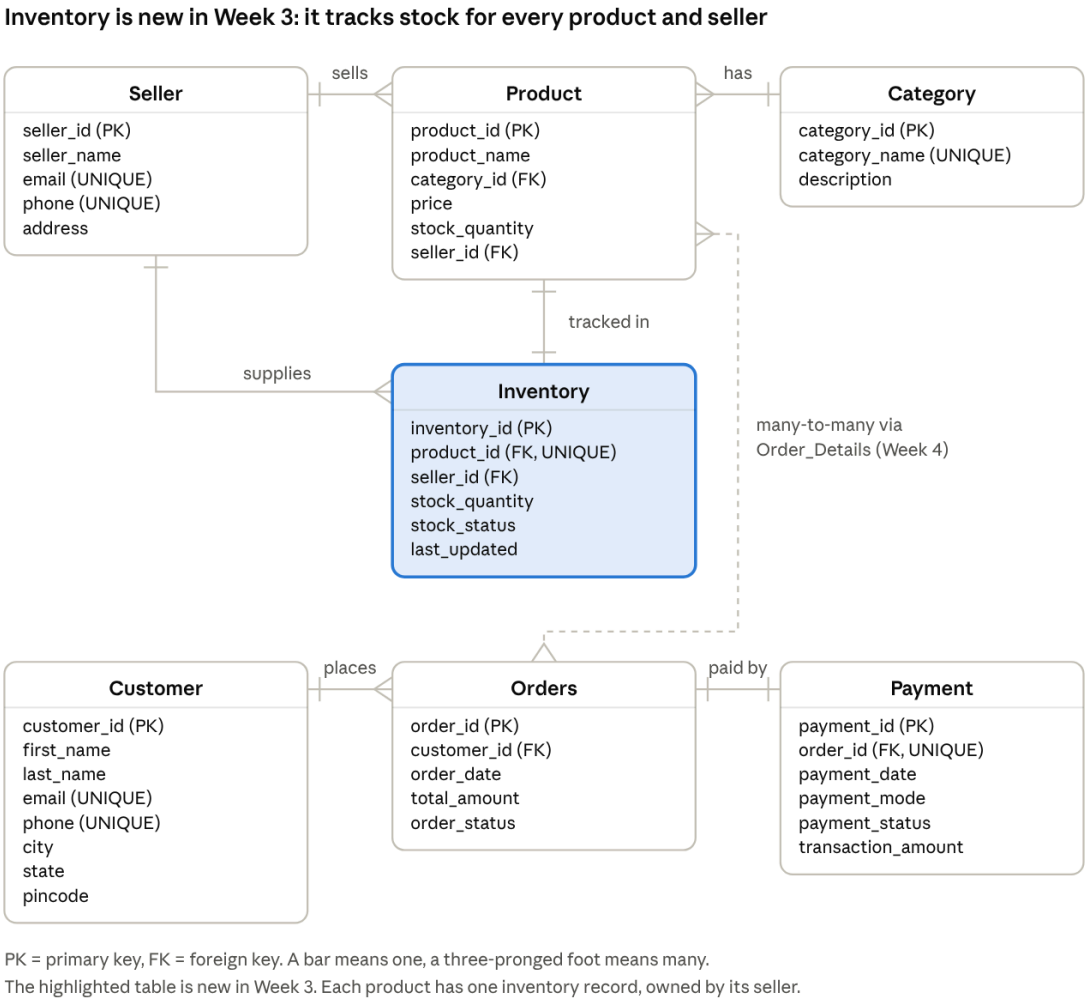
Relationship	Type	Linked by	Explanation
Seller to Product	One-to-Many	product.seller_id	One seller supplies many products. Each product has one seller.
Product to Inventory	One-to-One	inventory.product_id (UNIQUE)	Each product has at most one inventory record, which holds its current stock.
Seller to Inventory	One-to-Many	inventory.seller_id	One seller owns the inventory records of all the products it supplies.

Referential integrity rules

- A seller who still supplies products cannot be deleted (MySQL error 1451).
- An inventory record cannot be created for a product or seller that does not exist.
- Deleting a product also deletes its inventory record (ON DELETE CASCADE). Deleting only the inventory record leaves the product and its order history untouched.
- Stock cannot be negative, and a seller phone number must have exactly 10 digits.

2. ER Diagram

The updated ER diagram adds the inventory table to the entities from Tasks 1 and 2. The highlighted table is new in Week 3, and the crow's foot marks the many side of a relationship.



Updated ER diagram · 7 entities, 8 relationships

Each seller supplies many products and owns many inventory records. Each product has at most one inventory record, which holds its current stock. The relationships from Tasks 1 and 2 are unchanged.

3. Sample Seller and Inventory Data

3.1 Seller data

The first 10 sellers come from Task 1. Three new sellers (IDs 11 to 13) are added in Week 3. Later in the script the address of KitchenKart and the phone number and email of Urban Style Traders are updated.

Seller_ID	Seller_Name	Email	Phone	Address
1	TechWorld Retail	techworld@mail.com	9000000001	99 Brigade Road, Bengaluru, Karnataka
2	FashionHub	fashionhub@mail.com	9000000002	45 Linking Road, Mumbai, Maharashtra
3	HomeEssentials	homeess@mail.com	9000000003	8 Anna Salai, Chennai, Tamil Nadu
4	BookNest	booknest@mail.com	9000000004	21 Connaught Place, Delhi
5	SportsZone	sportszone@mail.com	9000000005	5 FC Road, Pune, Maharashtra
6	GadgetGalaxy	gadgetgx@mail.com	9000000006	33 Banjara Hills, Hyderabad, Telangana
7	KitchenKart	kitchenkart@mail.com	9000000007	17 RS Puram, Coimbatore, Tamil Nadu
8	BeautyBay	beautybay@mail.com	9000000008	9 Park Street, Kolkata, West Bengal
9	ToyTown	toytown@mail.com	9000000009	27 MI Road, Jaipur, Rajasthan
10	GreenLeaf Organics	greenleaf@mail.com	9000000010	14 West Masi Street, Madurai, Tamil Nadu
11	Urban Style Traders	urbanstyle@mail.com	9000000011	22 Residency Road, Bengaluru, Karnataka
12	Fresh Basket Foods	freshbasket@mail.com	9000000012	6 Race Course Road, Coimbatore, Tamil Nadu
13	Craft Corner	craftcorner@mail.com	9000000013	31 Bazaar Street, Jaipur, Rajasthan

Craft Corner is added without any products so that the product count per seller shows a seller with 0 products.

3.2 New products and seller assignment

Six new products are inserted with no seller. They are then assigned to sellers with UPDATE statements. Their IDs are generated automatically.

Product_Name	Category	Price	Stock_Quantity	Assigned seller
Denim Jacket	Fashion	1799.00	8	Urban Style Traders
Leather Wallet	Fashion	599.00	0	Urban Style Traders
Almonds 500g	Grocery	549.00	5	Fresh Basket Foods
Green Tea 100 Bags	Grocery	299.00	0	Fresh Basket Foods
Cotton Bedsheet	Home & Kitchen	899.00	60	HomeEssentials
Cricket Ball Pack	Sports	349.00	0	SportsZone

3.3 Inventory data

One inventory record is created for each of the 26 products with an INSERT ... SELECT statement. The stock_status label is set from the quantity with a CASE expression.

Inventory_ID	Product_Name	Seller	Stock_Quantity	Stock_Status
1	Redmi Note 13 5G	TechWorld Retail	50	In Stock
2	Men Cotton T-Shirt	FashionHub	200	In Stock
3	Non-stick Cookware Set	HomeEssentials	80	In Stock
4	Wings of Fire (Book)	BookNest	150	In Stock
5	Kashmir Willow Cricket Bat	SportsZone	40	In Stock
6	Wireless Headphones	GadgetGalaxy	120	In Stock
7	Mixer Grinder 750W	KitchenKart	60	In Stock
8	Face Wash 100ml	BeautyBay	300	In Stock
9	Building Blocks Set	ToyTown	90	In Stock
10	Organic Honey 500g	GreenLeaf Organics	110	In Stock
11	Bluetooth Speaker	GadgetGalaxy	70	In Stock
12	Smart Watch	TechWorld Retail	70	In Stock
13	Power Bank 20000mAh	GadgetGalaxy	100	In Stock

Inventory_ID	Product_Name	Seller	Stock_Quantity	Stock_Status
14	USB-C Fast Charger 65W	TechWorld Retail	85	In Stock
15	Women Kurti	FashionHub	120	In Stock
16	Running Shoes	FashionHub	60	In Stock
17	Steel Water Bottle 1L	HomeEssentials	150	In Stock
18	Atomic Habits (Book)	BookNest	90	In Stock
19	Yoga Mat	SportsZone	75	In Stock
20	Moisturizer 200ml	BeautyBay	140	In Stock
21	Denim Jacket	Urban Style Traders	8	Low Stock
22	Leather Wallet	Urban Style Traders	0	Out of Stock
23	Almonds 500g	Fresh Basket Foods	5	Low Stock
24	Green Tea 100 Bags	Fresh Basket Foods	0	Out of Stock
25	Cotton Bedsheet	HomeEssentials	60	In Stock
26	Cricket Ball Pack	SportsZone	0	Out of Stock

The CRUD section later changes this data. Leather Wallet is restocked from 0 to 50 units, 3 units of Almonds 500g are sold (5 to 2), and the discontinued Green Tea 100 Bags loses its inventory record. The analysis report below uses the data after these changes.

4. Inventory Analysis Report

This report presents the results of the seller and inventory queries in the SQL script, followed by the main observations.

4.1 Products supplied by each seller

Seller	Products supplied
BeautyBay	Face Wash 100ml, Moisturizer 200ml
BookNest	Atomic Habits (Book), Wings of Fire (Book)
FashionHub	Men Cotton T-Shirt, Running Shoes, Women Kurti

Seller	Products supplied
Fresh Basket Foods	Almonds 500g, Green Tea 100 Bags
GadgetGalaxy	Bluetooth Speaker, Power Bank 20000mAh, Wireless Headphones
GreenLeaf Organics	Organic Honey 500g
HomeEssentials	Cotton Bedsheet, Non-stick Cookware Set, Steel Water Bottle 1L
KitchenKart	Mixer Grinder 750W
SportsZone	Cricket Ball Pack, Kashmir Willow Cricket Bat, Yoga Mat
TechWorld Retail	Redmi Note 13 5G, Smart Watch, USB-C Fast Charger 65W
ToyTown	Building Blocks Set
Urban Style Traders	Denim Jacket, Leather Wallet

Craft Corner does not appear in this list because it supplies no products yet.

4.2 Number of products supplied by each seller

The query uses LEFT JOIN so that a seller with no products still appears with a count of 0.

Seller_Name	Total_Products
FashionHub	3
GadgetGalaxy	3
HomeEssentials	3
SportsZone	3
TechWorld Retail	3
BeautyBay	2
BookNest	2
Fresh Basket Foods	2
Urban Style Traders	2
GreenLeaf Organics	1
KitchenKart	1
ToyTown	1

Seller_Name	Total_Products
Craft Corner	0

4.3 Products with stock below 10

When this query runs, before the restock and the delete, five products have fewer than 10 units. Out-of-stock products are included because 0 is less than 10.

Product_Name	Stock_Quantity	Stock_Status
Cricket Ball Pack	0	Out of Stock
Green Tea 100 Bags	0	Out of Stock
Leather Wallet	0	Out of Stock
Almonds 500g	5	Low Stock
Denim Jacket	8	Low Stock

4.4 Stock updates and discontinued inventory

Product	Operation	Before	After
Leather Wallet	Restock	0 (Out of Stock)	50 (In Stock)
Almonds 500g	Sale of 3 units	5 (Low Stock)	2 (Low Stock)
Green Tea 100 Bags	Discontinued, inventory record deleted	0 (Out of Stock)	No inventory record

The status is refreshed by a follow-up UPDATE, and the last_updated time changes automatically when the quantity changes. The product row of Green Tea 100 Bags is kept, so its order history is not affected.

4.5 Seller-wise inventory summary

Seller_Name	Products_Listed	Total_Stock	Average_Stock
BeautyBay	2	440	220.00
FashionHub	3	380	126.67
GadgetGalaxy	3	290	96.67
HomeEssentials	3	290	96.67
BookNest	2	240	120.00
TechWorld Retail	3	205	68.33

Seller_Name	Products_Listed	Total_Stock	Average_Stock
SportsZone	3	115	38.33
GreenLeaf Organics	1	110	110.00
ToyTown	1	90	90.00
KitchenKart	1	60	60.00
Urban Style Traders	2	58	29.00
Fresh Basket Foods	1	2	2.00
Craft Corner	0	0	None

Fresh Basket Foods lists one product because the inventory record of Green Tea 100 Bags was deleted.

4.6 Stock availability report

Stock_Status	Total_Products	Total_Units
In Stock	22	2270
Low Stock	2	10
Out of Stock	1	0

The five products with the lowest stock are:

Product_Name	Seller	Stock_Quantity	Stock_Status
Cricket Ball Pack	SportsZone	0	Out of Stock
Almonds 500g	Fresh Basket Foods	2	Low Stock
Denim Jacket	Urban Style Traders	8	Low Stock
Kashmir Willow Cricket Bat	SportsZone	40	In Stock
Leather Wallet	Urban Style Traders	50	In Stock

4.7 Inventory figures

Measure	Result
Total available products (stock greater than 0)	24
Out-of-stock products	1 (Cricket Ball Pack, supplied by SportsZone)

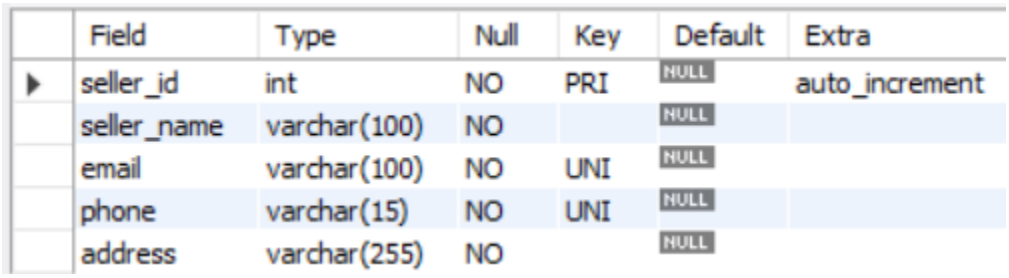
Measure	Result
Highest stocked product	Face Wash 100ml (BeautyBay), 300 units
Average inventory quantity	91.20 units per product

4.8 Key observations

- 24 of the 25 inventory records are available. Only Cricket Ball Pack is out of stock.
- Almonds 500g (2 units) and Denim Jacket (8 units) are low on stock and are the first products to reorder.
- BeautyBay holds the most stock, 440 of the 2280 units (about 19.3%), from only two products. Its Face Wash 100ml is the highest stocked product.
- The two newest sellers hold very little stock. Urban Style Traders has 58 units and Fresh Basket Foods has 2 units.
- Craft Corner has no products listed, so it needs products assigned before it can sell.
- The average of 91.20 units per product hides a wide spread, from 0 units for Cricket Ball Pack to 300 units for Face Wash 100ml.

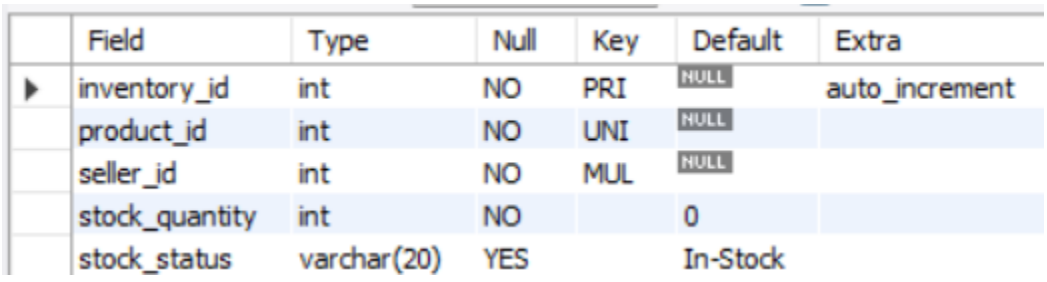
5. Query Output Screenshots

1. DESCRIBE seller



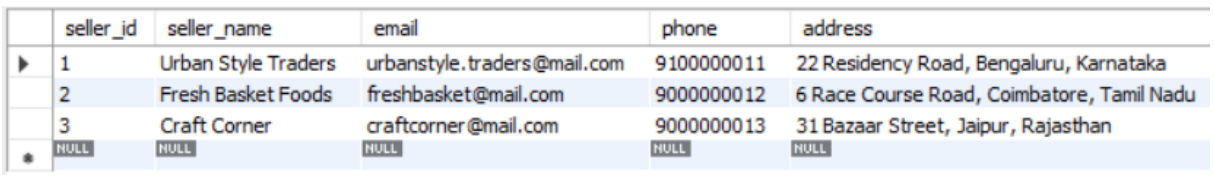
Field	Type	Null	Key	Default	Extra
seller_id	int	NO	PRI	NULL	auto_increment
seller_name	varchar(100)	NO		NULL	
email	varchar(100)	NO	UNI	NULL	
phone	varchar(15)	NO	UNI	NULL	
address	varchar(255)	NO		NULL	

2. DESCRIBE inventory



Field	Type	Null	Key	Default	Extra
inventory_id	int	NO	PRI	NULL	auto_increment
product_id	int	NO	UNI	NULL	
seller_id	int	NO	MUL	NULL	
stock_quantity	int	NO		0	
stock_status	varchar(20)	YES		In-Stock	

3. SELECT * FROM seller



seller_id	seller_name	email	phone	address
1	Urban Style Traders	urbanstyle.traders@mail.com	9100000011	22 Residency Road, Bengaluru, Karnataka
2	Fresh Basket Foods	freshbasket@mail.com	9000000012	6 Race Course Road, Coimbatore, Tamil Nadu
3	Craft Corner	craftcorner@mail.com	9000000013	31 Bazaar Street, Jaipur, Rajasthan
NULL	NULL	NULL	NULL	NULL

4. SELECT product_id, product_name FROM product

   WHERE seller_id IS NULL;

	product_id	product_name
▶	6	Cotton Bedsheet
	7	Cricket Ball Pack
*	NULL	NULL

5. SELECT s.seller_name, p.product_id, p.product_name, p.price

   FROM seller s

   JOIN product p ON s.seller_id = p.seller_id

   ORDER BY s.seller_name, p.product_name;

	seller_name	product_id	product_name	price
▶	Fresh Basket Foods	4	Almonds 500g	549.00
	Fresh Basket Foods	5	Green Tea 100 Bags	299.00
	Urban Style Traders	2	Denim Jacket	1799.00
	Urban Style Traders	3	Leather Wallet	599.00

6. SELECT s.seller_id, s.seller_name, COUNT(p.product_id) AS total_products

   FROM seller s

   LEFT JOIN product p ON s.seller_id = p.seller_id

   GROUP BY s.seller_id, s.seller_name

   ORDER BY total_products DESC, s.seller_name;

	seller_id	seller_name	total_products
▶	2	Fresh Basket Foods	2
	1	Urban Style Traders	2
	3	Craft Corner	0

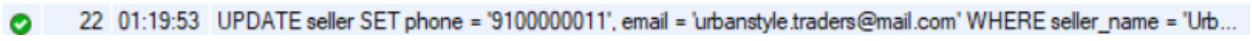
7. UPDATE seller SET address = '18 Avinashi Road, Coimbatore, Tamil Nadu'

   WHERE seller_name = 'KitchenKart';



UPDATE seller SET phone = '9100000011', email = 'urbanstyle.traders@mail.com'

WHERE seller_name = 'Urban Style Traders';



8. SELECT seller_id, seller_name, email, phone, address FROM seller

   WHERE seller_name IN ('KitchenKart', 'Urban Style Traders');

seller_id	seller_name	email	phone	address
1	Urban Style Traders	urbanstyle.traders@mail.com	9100000011	22 Residency Road, Bengaluru, Karnataka
NULL	NULL	NULL	NULL	NULL

9. SELECT * FROM inventory;

inventory_id	product_id	seller_id	stock_quantity	stock_status
1	2	1	8	In-Stock
2	3	1	50	In-Stock
3	4	2	2	In-Stock
4	5	2	0	In-Stock
NULL	NULL	NULL	NULL	NULL

10. Available products

product_id	product_name	stock_quantity	stock_status
4	Almonds 500g	2	In-Stock
2	Denim Jacket	8	In-Stock
3	Leather Wallet	50	In-Stock

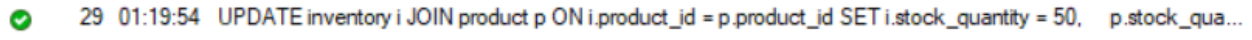
11. Out-of-stock products

product_id	product_name	seller_name	stock_status
5	Green Tea 100 Bags	Fresh Basket Foods	In-Stock

12. Stock below 10

product_id	product_name	stock_quantity	stock_status
5	Green Tea 100 Bags	0	In-Stock
4	Almonds 500g	2	In-Stock
2	Denim Jacket	8	In-Stock

13. Stock updates



14. DELETE of discontinued inventory

```
43  01:38:13  DELETE FROM inventory WHERE product_id = (SELECT product_id FROM product WHERE product_name ...
```

15. Seller-wise product report

seller_name	product_name	category_name	price	stock_quantity	stock_status
Fresh Basket Foods	Almonds 500g	Grocery	549.00	2	In-Stock
Urban Style Traders	Denim Jacket	Fashion	1799.00	8	In-Stock
Urban Style Traders	Leather Wallet	Fashion	599.00	50	In-Stock

16. Seller-wise summary

stock_status	total_products	total_units
In-Stock	3	60

17. Stock availability report

product_name	seller_name	stock_quantity	stock_status
Almonds 500g	Fresh Basket Foods	2	In-Stock
Denim Jacket	Urban Style Traders	8	In-Stock
Leather Wallet	Urban Style Traders	50	In-Stock

18. Total available products

total_available_products
3

19. Out-of-stock products

product_id	product_name	seller_name

20. Highest stocked product

product_id	product_name	seller_name	stock_quantity
3	Leather Wallet	Urban Style Traders	50

21. Average inventory quantity

	average_inventory_quantity
▶	20.00

	seller_name	average_stock
▶	Urban Style Traders	29.00
	Fresh Basket Foods	2.00

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II BCA Batch 1